

AI-Generated Image Detection for Etsy Product Listings

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Abstract— The widespread adoption of AI image generation tools such as Midjourney, DALL-E, and Stable Diffusion has fundamentally transformed digital content creation, introducing significant challenges for e-commerce platforms like Etsy, where product authenticity is central to buyer trust. When sellers substitute genuine product photographs with synthetically generated imagery, buyer confidence is undermined and marketplace integrity is compromised. At the scale of millions of listings, manual review is neither practical nor economically viable, making automated detection an operational necessity. This paper presents a robust multi-model ensemble pipeline for binary classification of product listing images as either authentic (label 0) or AI-generated (label 1), evaluated using the F1 score to balance precision and recall. The proposed system combines four complementary architectures EfficientNet-B4, Vision Transformer (ViT-B/16), ConvNeXt-Base, and a frequency-domain classifier leveraging DCT and DWT features to capture both spatial texture artefacts and spectral patterns invisible in pixel space. Models are trained with 5-fold Stratified K-Fold cross-validation, progressive backbone unfreezing, differential learning rates, Label Smoothing Cross-Entropy loss, and Mixup/CutMix augmentation. Inference employs 10-step Test-Time Augmentation, temperature scaling for probability calibration, and weighted ensemble combination with weights optimised via grid search on Out-of-Fold predictions. A decision threshold of 0.576 further refines the precision-recall trade-off. The final pipeline achieves an OOF F1 of 0.9270, substantially outperforming any single-model baseline.

Keywords— AI-Generated Image Detection, Binary Image Classification, Transfer Learning, Ensemble Learning, Vision Transformer, E-Commerce Content Moderation.

I. INTRODUCTION

The rapid advancement of generative artificial intelligence has made it possible to produce photorealistic product imagery at negligible cost. Platforms like Etsy, whose value proposition depends on handcrafted and authentic goods, face a growing risk of sellers substituting genuine product photographs with synthetic images generated by tools such as Midjourney, DALL-E, or Stable Diffusion. Such misrepresentation damages buyer trust and undermines the integrity of the marketplace. Manual review at the scale of millions of listings is neither feasible nor economically sustainable, making automated detection an operational necessity.

This work frames the problem as a supervised binary classification task. Each listing image is assigned a label of 1 (AI-generated) or 0 (authentic), and performance is evaluated using the F1 score, which balances precision and recall. The F1 metric is particularly appropriate here because both false positives, authentic images incorrectly removed, and false negatives, AI-generated images that pass

undetected, carry meaningful business consequences [1]. The detection task is non-trivial for several reasons. First, AI-generated images are highly diverse in style, subject matter, and generation method; a model trained primarily on one generator's outputs may not generalise to others. Second, many AI-generated product images are deliberately styled to appear authentic, with realistic lighting, backgrounds, and textures. Third, some authentic product photographs may exhibit characteristics superficially similar to AI-generated images due to heavy post-processing or studio photography techniques [16].

To address these challenges, we design a multi-architecture ensemble that integrates the complementary strengths of convolutional networks, a vision transformer, and a frequency-domain classifier. The key contributions of this work are:

- A four-model ensemble combining EfficientNet-B4, ViT-B/16, ConvNeXt-Base, and a DCT/DWT frequency-domain classifier.
- A two-stage progressive fine-tuning strategy with differential learning rates and cosine annealing with warm restarts.
- Temperature scaling and OOF-optimised ensemble weighting for improved probability calibration.
- 10-step Test-Time Augmentation (TTA) at inference for more stable and robust predictions.

The remainder of this paper first reviews relevant literature on AI-generated image detection before detailing the proposed ensemble pipeline, including dataset preparation, model architectures, and training design. Experimental results and ablation studies are then presented, followed by concluding remarks and directions for future work.

II. RELATED WORK

The detection of AI-generated images has evolved rapidly alongside advances in generative models. The relevant literature spans four key themes: CNN-based artifact detection approaches, Vision Transformer and large-scale encoder methods, frequency-domain analysis, and generalisation and robustness strategies. This work draws on all four themes to construct a robust ensemble classifier for binary AI-generated image detection in an e-commerce product image context.

A. AI-Generated Image Detection Methods

Early approaches to AI-generated image detection primarily relied on Convolutional Neural Networks (CNNs) to learn discriminative artifact features. CNN-based models demonstrated strong performance by capturing subtle inconsistencies in texture, color, and background patterns introduced during image synthesis. For instance, CNN

architectures trained on synthetic datasets such as CIFAKE achieved high classification accuracy while revealing that models often rely on background imperfections rather than semantic content Bird et al [1].

Lightweight CNN models have also been proposed to improve efficiency while maintaining competitive accuracy, demonstrating that even shallow architectures can effectively distinguish synthetic from real images Ladevi et al [2]. Furthermore, studies using adapted ResNet-based architectures showed that deep CNNs can generalise across multiple generative models when trained on diverse datasets Saskaro et al [3]. However, these approaches often struggle when exposed to unseen generative techniques, highlighting limitations in generalisation.

B. Frequency-Domain and Fine-Grained Feature Approaches

To address the limitations of spatial-domain methods, several studies have explored frequency-domain representations as a source of more generalisable forgery traces. Frequency-based approaches leverage transformations such as Fourier Transform and Discrete Wavelet Transform (DWT) to reveal hidden artifacts introduced during image generation. Recent work has demonstrated that combining spatial and frequency features significantly improves detection robustness. For example, dual-frequency frameworks integrate multiple frequency bands to capture both global and local inconsistencies, leading to improved detection accuracy across diverse datasets Yan et al [5].

Additionally, preprocessing techniques such as frequency filtering and cross-difference operations have been shown to enhance CNN performance by emphasizing subtle noise patterns that are otherwise invisible in the spatial domain Metwally & Ghantous [4]. Beyond frequency-based approaches, color space analysis has also been explored as a discriminative feature, where differences in RGB, HSV, and YCbCr channel distributions between real and synthetic images are leveraged to improve detection accuracy, achieving performance above 99% when combined with attention-based deep learning models Mo et al [7]. These findings highlight the importance of multi-domain feature extraction for robust AI-image detection.

C. Vision Transformers and Large-Scale Encoder Methods

With the emergence of transformer architectures, Vision Transformers (ViTs) have become a powerful alternative to CNNs for image classification tasks. Unlike CNNs, ViTs model global dependencies through self-attention, enabling them to capture long-range relationships within images. Recent studies demonstrate that ViT-based models outperform traditional CNNs in detecting AI-generated images, particularly when fine-tuned with data augmentation and pretrained weights Lamichhane [8]. Hybrid approaches combining CNNs and transformers have also shown improved performance by leveraging both local feature extraction and global contextual understanding Arshed et al [9]. These methods are especially effective in handling diverse generative models, making them suitable for real-world applications where image distributions vary significantly.

D. Ensemble and Multi-Model Detection

Given the diversity of generative models, single-model approaches often fail to generalise across different data

distributions. Ensemble-based methods have been proposed to address this limitation by combining multiple models with complementary strengths. A notable approach is the gated expert CNN framework, where multiple specialised CNN models are combined using a gating mechanism that dynamically weights their outputs. This method improves robustness across different generators while maintaining computational efficiency Saskaro et al [3].

Similarly, recent work on detection and attribution systems demonstrates that ensemble learning can enhance both classification accuracy and model interpretability, particularly in complex real-world scenarios Namani et al [10]. These findings strongly motivate the use of ensemble strategies in AI-generated image detection tasks.

E. Generalisation, Diffusion Models, and Robustness

A major challenge in AI-generated image detection is generalisation to unseen generative models, particularly diffusion-based approaches. Studies show that detectors trained on GAN-generated images often fail when applied to diffusion-generated images due to differences in artifact distributions Zhang et al [11]. To address this, recent methods focus on learning generalisable features rather than model-specific artifacts. For example, pixel-level preprocessing techniques have been proposed to reduce semantic bias and improve cross-model performance Zhou et al [12]. Similarly, semantic-guided regeneration approaches decouple image content from generative artifacts, enabling more robust detection under limited data conditions Yu et al [13].

Dogoulis et al. [6] highlighted the challenge of cross-concept generalisation, showing that detectors trained on one semantic category (e.g., human faces) perform poorly when evaluated on different concepts (e.g., animals), and proposed a quality-based sampling strategy that significantly improves robustness across diverse image domains. Emerging approaches such as hybrid classical-quantum neural networks further aim to improve generalisation by enhancing feature representation capabilities Xu et al [15]. Park et al. [14] confirmed that methods trained without these augmentations (e.g., DIF, ViT) exhibit sharp performance degradation under JPEG compression and Gaussian blur in real-world scenarios. Data augmentation techniques and multi-domain feature extraction have been shown to significantly improve model stability under such conditions, making them essential components of modern detection pipelines.

In summary, the literature on AI-generated image detection demonstrates significant progress through CNN-based artifact learning, frequency-domain analysis, transformer-based models, and ensemble strategies. While each approach offers distinct advantages, no single method consistently achieves strong generalisation across diverse generative models and real-world conditions. In particular, challenges remain in handling unseen generators, diffusion-based images, and post-processing distortions such as compression and noise. These limitations highlight the need for more robust and adaptable detection frameworks. Motivated by these gaps, this work adopts an ensemble-based approach that integrates complementary feature representations to enhance generalisation and reliability in practical e-commerce image classification scenarios.

TABLE I. Summary of Related Work on AI-Generated Image Detection

Ref.	Method	Key Contribution	Limitation
[1]	CNN on CIFAKE dataset	High accuracy detecting AI images; models rely on background imperfections	Limited to specific dataset; poor semantic generalisation
[2]	Lightweight CNN	Efficient shallow architecture distinguishing real vs synthetic	Reduced capacity limits performance on complex generators
[3]	ResNet + Gated Expert CNN Ensemble	Cross-generator generalisation; dynamic model weighting	Struggles with unseen generative techniques
[4]	Frequency filtering + cross-difference CNN	Highlights subtle noise patterns invisible in spatial domain	Spatial features still dominant; limited frequency depth
[5]	Dual-frequency framework	Captures global and local inconsistencies across frequency bands	Complexity increases; limited real-world e-commerce testing
[6]	Quality-based sampling strategy	Addresses cross-concept generalisation gap; improves robustness across diverse semantic domains	Trained on specific categories; sampling strategy adds overhead
[7]	Color space analysis (RGB, HSV, YCbCr) + attention DL	Above 99% accuracy using multi-channel distributions	Controlled setting; may not generalise to product images
[8]	Fine-tuned ViT with augmentation	ViT outperforms CNNs on AI-generated image detection	Computationally expensive; needs strong pretrained weights
[9]	Hybrid CNN-Transformer	Combines local and global feature extraction	Higher complexity; limited evaluation on diverse generators
[10]	Ensemble detection and attribution	Improves accuracy and interpretability in real-world scenarios	Attribution overhead; single-domain focus
[11]	GAN vs diffusion detector analysis	Identifies artifact distribution shift between GAN and diffusion	GAN-trained detectors fail on diffusion images

[12]	Pixel-level preprocessing	Reduces semantic bias; improves cross-model performance	Preprocessing sensitivity; may discard useful features
[13]	Semantic-guided regeneration	Decouples content from generative artifacts	Limited data conditions; regeneration overhead
[14]	Augmentation robustness analysis	JPEG/blur degradation confirmed without augmentation	Augmentation-dependent; not a detection model itself
[15]	Hybrid classical-quantum neural network	Enhanced feature representation for generalisation	Quantum computing accessibility limits practical deployment

III. METHODOLOGY

This section describes the complete pipeline for AI-generated image detection on the Etsy product listing dataset, organised across six components: dataset description, preprocessing and augmentation, model architectures, training strategy, test-time augmentation, and ensemble calibration. The design is motivated by the need for complementary representations. Convolutional and transformer-based models detect spatial texture artefacts and semantic inconsistencies, while frequency-domain analysis exposes spectral irregularities invisible in pixel space. The ensemble leverages the distinct inductive biases of each component to improve generalisation. All models were implemented in PyTorch using the timm library, Albumentations for augmentation, and scikit-learn for the frequency-domain classifier, with GPU-based mixed-precision training throughout. The complete pipeline is illustrated in Fig. 1.

The methodology presented in this section reflects the final iteration of a three-stage development process. The first iteration achieved a test-set F1 of 0.9057 by establishing the core pipeline with EfficientNet-B4 and ViT-B/16 trained under 5-fold cross-validation, basic temperature scaling, and single-scale TTA. The second iteration improved performance to 0.9180 through two key enhancements: a local checkpoint system that cached trained model weights to ensure reproducible OOF probability estimates across runs, and the introduction of multi-scale TTA, which ran inference at a secondary resolution for each spatial model and averaged the results with primary-scale predictions to reduce resolution-dependent variance. The final iteration achieved the submitted test-set F1 of 0.9270 by incorporating a more exhaustive ensemble weight search, a finer threshold sweep over [0.28, 0.72] in steps of 0.002, and improved post-processing calibration, collectively identifying an optimal decision boundary of 0.576. All methodology described in the following subsections refers to this final pipeline.



Fig. 1. End-to-end pipeline for AI-generated image detection on Etsy product listings.

A. Dataset

The dataset was provided by Etsy for this challenge and consists of 6,858 product listing images from real-world e-commerce scenarios. The training split contains 4,800 labelled images with a near-balanced class distribution: 2,485 authentic images (label 0) and 2,315 AI-generated images (label 1). The test split contains 2,058 unlabelled images for which predictions were submitted for evaluation. The dataset reflects the natural diversity of product imagery, including variations in lighting, background, composition, and image quality, as well as synthetic images produced by multiple contemporary generative models.

B. Data Preprocessing and Augmentation

All images were loaded using OpenCV and converted from BGR to RGB format to align with the colour ordering expected by pretrained deep learning models. A comprehensive augmentation pipeline was constructed using the Albumentations library. Spatial augmentations included horizontal flipping ($p=0.5$), vertical flipping ($p=0.2$), random 90-degree rotations ($p=0.3$), and shift-scale-rotate transformations ($p=0.4$). Photometric augmentations brightness, contrast, saturation, and hue adjustments ($p=0.5$) accounted for lighting diversity. CLAHE ($p=0.3$) was applied for local contrast enhancement, Gaussian noise ($p=0.3$) to simulate sensor distortions, and coarse dropout ($p=0.2$) to encourage distributed feature learning. All images were normalised using ImageNet statistics. Input resolutions were 300×300 for EfficientNet-B4 and ConvNeXt-Base, and 224×224 for ViT-B/16.

C. Model Architectures

To capture diverse feature representations, the pipeline combines three spatial deep learning models with a handcrafted frequency-domain classifier.

Spatial Models: A unified AIDetector architecture was implemented using the timm library, with pretrained backbones followed by a custom head comprising layer normalisation, dropout, and fully connected layers with GELU activation. Three backbone architectures were employed: (i) EfficientNet-B4, selected for its compound scaling and strong performance on fine-grained visual detail [21]; (ii) ViT-B/16, which uses multi-head self-attention to capture long-range semantic dependencies [17]; and (iii)

ConvNeXt-Base, a modernised convolutional architecture designed to match transformer performance while retaining the inductive biases of CNNs [18]. All models were initialised with ImageNet-pretrained weights.

Frequency-Domain Classifier: A dedicated frequency-domain classifier was designed to capture spectral and textural artifacts characteristic of AI-generated imagery. Features were extracted from a 128×128 grayscale representation spanning seven groups: FFT radial spectrum statistics; DCT high-frequency residuals from the top-right quadrant; DWT detail coefficients using db4 wavelets at two decomposition levels; Gaussian noise residual statistics; JPEG block artifact variance across 8×8 boundaries; YCrCb chrominance noise statistics; and Local Binary Pattern (LBP) histogram features for micro-texture characterisation. Channel-wise RGB colour statistics were also included. The complete feature vector was standardised and fed into a Gradient Boosting classifier (scikit-learn), targeting spectral and textural irregularities introduced during diffusion model synthesis.

D. Training Strategy

All spatial models were trained using 5-fold Stratified K-Fold cross-validation to preserve class distribution and yield reliable OOF probability estimates. Training followed a two-stage progressive fine-tuning schedule. In Stage 1 (epochs 1–5), the backbone was frozen and only the classification head was optimised, preventing disruption of pretrained ImageNet features. At epoch 6, the backbone was unfrozen and end-to-end fine-tuning began with differential learning rates: 3e-6 for the backbone and 2e-4 for the head. The Cosine Annealing with Warm Restarts scheduler was reset dynamically upon unfreezing, with T_0 set to the remaining epoch count per fold. The AdamW optimiser was used throughout with mixed-precision training and gradient clipping for efficiency and stability. The loss function was Label Smoothing Cross-Entropy with custom Focal Loss (smoothing=0.03) along with class-balanced sample weighting $w_c = N / (2 \times N_c)$. Mixup ($\alpha=0.2$, $p=0.5$) [19] and CutMix [20] augmentations were applied to improve generalisation on ambiguous examples.

E. Test-Time Augmentation

At inference time, a two-stage Test-Time Augmentation strategy was applied to all spatial models. In the first stage, 10 augmented versions of each image were generated using random horizontal flips, vertical flips, and 90-degree rotations at the primary training resolution. In the second stage, five additional inference rounds were performed at a secondary resolution 256px for EfficientNet-B4 and ConvNeXt-Base, and 224px for ViT-B/16 using identical augmentations. Primary-scale and secondary-scale softmax probabilities were blended with equal weighting (0.5 each) before ensemble combination, reducing both augmentation variance and resolution-dependent prediction bias.

F. Calibration and Ensemble Strategy

Temperature scaling was applied to each model's OOF predictions to correct for overconfidence. The optimal temperature T was found by minimising negative log-likelihood over a grid search on validation data, with T searched over [0.4, 2.2] in steps of 0.1. Ensemble weights were then determined by grid search over calibrated OOF probabilities to maximise OOF F1. The resulting weights were: EfficientNet-B4 (0.15), ViT-B/16 (0.40), ConvNeXt-Base (0.20), and the frequency classifier (0.25). The classification threshold was tuned from 0.5 to 0.576 using a fine-grained

search over [0.28, 0.72] in steps of 0.002, selected to maximise validation F1.

IV. EXPERIMENTS AND RESULTS

This section presents the experimental results of the proposed pipeline. We first report individual model performance across the 5-fold cross-validation scheme, followed by a comparison of ensemble configurations and threshold settings. Finally, we analyse the contribution of key training design choices to overall performance.

A. Individual Model Performance

Table II reports the mean OOF F1 scores for all four models evaluated under the 5-fold Stratified K-Fold cross-validation scheme, comparing spatial deep learning architectures against the frequency-domain classifier.

TABLE II. Individual Model Performance Summary

Model	Architecture Type	Input Size	Mean OOF F1
EfficientNet-B4	CNN (Compound Scaling)	300×300	0.7884
ViT-B/16	Vision Transformer	224×224	0.8853
ConvNeXt-Base	Modern CNN	288×288	0.8785
Frequency Classifier	DCT / DWT + Gradient Boosting	128×128	0.9145

ViT-B/16 outperformed all other individual models, achieving the highest mean OOF F1 of 0.8853, compared to 0.8785 for ConvNeXt-Base, 0.9145 for the frequency classifier, and 0.7884 for EfficientNet-B4. This aligns with the expectation that transformer-based global attention is well-suited to detecting long-range semantic inconsistencies present in AI-generated images. A clear improvement was observed in both CNN and transformer models following backbone unfreezing, with ViT showing the most pronounced gain, indicating effective adaptation of its pretrained attention heads to the detection task.

B. Ensemble Performance

Table III summarises the progressive improvement achieved through ensemble construction and threshold optimisation.

TABLE III. Ensemble Performance Comparison

Configuration	OOF F1	Gain over Best Single Model
EfficientNet-B4 alone	0.7884	—
ConvNeXt-Base alone	0.8785	—
ViT-B/16 alone	0.8853	—
4-model ensemble (threshold = 0.500)	0.9250	+0.0397
4-model ensemble (threshold = 0.576)	0.9270	+0.0417

The four-model ensemble significantly outperformed all individual models, with an improvement of approximately +0.0397 F1 over the best standalone model (ViT, OOF F1 = 0.8853). Threshold tuning from 0.500 to 0.576 yielded a

further gain of +0.0020 in OOF F1, bringing the final score to 0.9270. The consistency across all five folds confirms that the OOF estimate is a reliable indicator of generalisation performance and that the pipeline did not overfit to the training data.

C. Impact of Training Design Choices

Several design decisions contributed meaningfully to overall performance. Progressive fine-tuning stabilised early training and prevented disruption of pretrained ImageNet features. Label smoothing reduced overconfident predictions, directly benefiting ensemble calibration. Mixup and CutMix augmentation improved performance on borderline examples by exposing the model to interpolated image-label pairs. Temperature scaling was particularly important for ViT, whose raw predictions were notably overconfident; calibration improved the reliability of its probability estimates and its effective weight in the ensemble. Collectively, these strategies produced a well-generalised and stably calibrated final pipeline.

V. CONCLUSION

This paper presented a multi-architecture ensemble framework for detecting AI-generated product images in Etsy listings. By integrating EfficientNet-B4, ViT-B/16, ConvNeXt-Base, and a frequency-domain classifier based on DCT and DWT features, the proposed system leverages complementary spatial and spectral representations. The pipeline achieved an OOF F1 score of 0.9270, substantially outperforming any single-model baseline. Among individual architectures, ViT-B/16 emerged as the strongest performer, highlighting the importance of global context modelling for synthetic image detection. The frequency-domain classifier provided complementary information that contributed to ensemble gains beyond what spatial models alone could achieve. Key findings confirm that training design choices, including progressive fine-tuning, label smoothing, calibration, and threshold optimisation are critical components of a high-performing detection pipeline. Future directions include exploring larger transformer architectures such as ViT-L and Swin Transformer, applying attention visualisation for improved interpretability, and developing robustness strategies for adversarially crafted or heavily post-processed synthetic images. As generative models continue to evolve in realism and diversity, adaptive detection frameworks will be essential for maintaining reliable platform integrity.

CONTRIBUTION

Rahul Satheesan Nair led the implementation of the EfficientNet-B4 and ViT-B/16 training pipeline, including the two-stage progressive backbone unfreezing strategy, differential learning rate scheduling, and 5-fold Stratified K-Fold cross-validation framework. He was primarily responsible for the frequency-domain feature extraction pipeline using DCT, DWT, and FFT radial spectrum analysis, and the Gradient Boosting classifier integration. Sooryakiran Vinod led the ConvNeXt-Base model development, multi-scale Test-Time Augmentation implementation, and the ensemble calibration strategy including temperature scaling and OOF-based weight optimisation. He was primarily responsible for threshold tuning, submission generation, and the overall pipeline integration across all four model components. Both authors jointly contributed to data preprocessing, augmentation pipeline design using Albumentations, result analysis, and the preparation of this report.

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